Research article

Temporal Change in Modeled Connectivity Among Southern African Cheetah (*Acinonyx jubatus jubatus*) Range Cores, 2012–2024

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Abstract

The southern African cheetah (*Acinonyx jubatus jubatus*) has lost much of its historical range, and most of its mapped 2010–2016 range lies outside protected areas. Habitat suitability in the region has been widely studied, but change over time in connectivity between known range areas has not been measured. This study modeled potential structural connectivity among 23 range cores from the 2010–2016 range assessment. The cores were held fixed across four snapshots (2012, 2016, 2020, and 2024), so every change reported reflects the landscape rather than the cheetahs. A resistance surface built from human pressure, vegetation, slope, and veterinary fences was used to model least-cost paths and circuit-theory current, and its assumptions were varied across nine surfaces. Across all 253 core pairs, the median modeled cost of moving between cores changed by only −1.2 percent; 161 pairs became easier and 92 became harder. Outside the cores, 23 percent of the land above the 80th percentile of modeled current was protected, against 31 percent of the comparison area, although protection rose to 45 percent in the top 1 percent. Of 45 selected links, 32 remained stable under the sensitivity tests and 13 were uncertain. In GPS tracking data from nine free-ranging cheetahs, daily moves ended on lower modeled resistance than comparable alternative moves for seven of the nine animals, a modest lean carried mostly by roads. These results suggest that change was mixed rather than a broad decline, and that land outside protected areas carries much of the modeled connectivity.

Keywords: circuit theory; least-cost path; resistance surface; protected area coverage; monitoring prioritisation; veterinary fences

1 Introduction

The southern African cheetah has had its range substantially reduced, and its mapped 2010–2016 range is estimated to lie mostly on unprotected land (Weise et al. 2017b). Cheetah range on protected land is governed by centralized park-management initiatives (Buk et al. 2018). Unprotected land poses a different type of challenge, with landowners, local communities, and governmental entities all involved. This paper examines landscape connectivity change between 2012 and 2024. The core range of cheetahs was fixed at the 2010–2016 baseline, so modeled changes in permeability reflect landscape change rather than change in the core range itself. Throughout, potential structural connectivity refers to modeled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a location of concentrated modeled current or a link-level bottleneck, not an observed crossing.

Previous research in the region has predominantly focused on habitat suitability, a topic that has been widely studied. The objective of this study was not to develop an additional suitability model, but rather to identify and evaluate candidate spatial "pinch points" (i.e., constricted linkages) among core areas. The study most comparable to the present work is a recent multi-species connectivity assessment of the Kavango–Zambezi Transfrontier Conservation Area (Sousa et al. 2026); however, that analysis excluded the cheetah because of insufficient data. Data limitations were therefore considered and mitigated to the extent possible in this study, and they provide both the primary justification for this paper and its main constraint. The approach combines range cores fixed at a historical baseline, repeated landscape snapshots from 2012 to 2024, and an independent check against GPS tracking data.

Two primary connectivity approaches are commonly used to model these kinds of frameworks: least-cost connectivity and circuit-theoretic connectivity (Adriaensen et al. 2003; McRae et al. 2008). The least-cost approach links core areas by selecting the route with the lowest cumulative "cost," meaning the path of least resistance between cores, based on the resistance assigned to each landscape factor. In contrast, the circuit-theoretic approach represents the landscape as a resistor network, sends a "current" through it, and highlights redundancy across the broader landscape, relying on the same resistance values. A key limitation of both methods is that a resistance value must first be assigned to every cell (1 km in this study). Zeller et al. (2012) demonstrated that even reasonable alternative assumptions about resistance values can influence results more strongly than the choice of connectivity method itself. For that reason, rather than treating one resistance surface as definitive, the weights were varied along two axes (Section 2.4). This reduces how much the results depend on any one choice of weights.

The objectives were to (1) measure how modeled connectivity between fixed range cores changed from 2012 to 2024, and (2) identify which connections stayed robust under different model assumptions and how much of the connecting land is protected. Four hypotheses follow from these objectives.

Temporal-change hypothesis. Temporal change in connectivity is expected to be strongest among a minority of links, and the links that are most important to the network are expected to experience the greatest loss between 2012 and 2024. The importance of a link is measured by its betweenness in the network, which is the number of shortest paths between cores that pass through it. A link with a high betweenness is therefore more important to the network than a link with a low betweenness. This was compared among the 45 selected links. This hypothesis would be refuted if loss were spread roughly evenly across links, or fell mainly on links with low betweenness.

Protected-area-coverage hypothesis. High-value connectivity areas are those with the highest modeled current. These areas are expected to be located disproportionately outside formal protected areas. This would be refuted if the unprotected share of high-current area were not greater than in the comparison area, or if the result did not hold across current thresholds.

Link-stability hypothesis. The stability of a link is defined as its ability to maintain connectivity under varying model assumptions. Some links may be more sensitive to changes in the landscape because of their location or the surrounding land use (e.g., a link surrounded by agricultural land may be more sensitive to land-use change than a link surrounded by protected areas). A small group of links is expected to remain important under all resistance scenarios, while other links are expected to be sensitive to the model changes. This would be refuted if agreement between scenarios were uniformly high, meaning the sensitivity tests added nothing, or uniformly low, meaning no stable group exists.

Monitoring-priority hypothesis. The active monitoring of cheetahs in the wild is as challenging as it is important, and the areas chosen for monitoring determine how efficiently conservation funding is used. The highest-priority places to monitor are expected to be where network importance overlaps with old or sparse cheetah records and fast landscape change.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range (Weise et al. 2017b) plus a 100-km buffer, so that modeled paths could route around the edge of the range rather than being cut off by it. All layers were aligned to a fixed equal-area projection (Africa Albers Equal Area Conic) and a 1 km snap grid, so every raster was directly comparable cell for cell. Cores were derived from the 2010–2016 cheetah density map and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs (Fig. 1). A 1-km working resolution was used because it represented a compromise between the coarser 10-km occurrence baseline and the finer environmental covariates, which ranged from 30 m to 1 km. This resolution was fine enough to identify meaningful points in the landscape while avoiding the implication of a level of accuracy that the occurrence evidence alone could not provide.

2.2 Range cores

A core area was defined as a contiguous patch of cells with a density of ≥ 0.51 cheetahs per 100 km^2 (Weise et al. 2017a). This value was where the planned top-10, 20, and 30 percent density rules all converged, because it covers a large, uniform part of the density map. The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the absence of a comparable range reassessment. The minimum core size was 500 km^2, roughly the home range of a territorial male or a female cheetah (Melzheimer et al. 2020). The protocol had planned three minimum sizes (2,000, 1,000, and 500 km^2); only the 500-km^2 definition was carried through the analysis.

2.3 Covariates and their temporal treatment

The temporal change in resistance came from three continuous, regularly updated layers: human development from GHSL, nighttime lights from VIIRS, and vegetation continuous fields (VCF) from MODIS (Table 1). Categorical land cover was used for masking and description only, because annual categorical maps can switch classes between years for reasons unrelated to overall landscape change.

Static covariates (roads, livestock, terrain, and fences) were also used. These did not contribute *temporal* signal to the resistance surface; however, they did contribute *spatial* signal to it. Hydrography and protected areas were used only to describe the landscape and to report protected-area coverage.

2.4 Resistance scenarios

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The main surface had a weight of 60% for human pressure, 20% for vegetation, and 20% for terrain. Within human pressure, built-up land, roads, and livestock each received 30%, while nighttime lights received 10%. Fence data were incorporated into the resistance model: cells without a fence kept their resistance, and fence cells were multiplied by 25. Fences were not treated as complete barriers, because they do not fully block large carnivores (Cozzi et al. 2013). Slopes of 10 degrees or less were given a resistance of 1, and slopes between 10 and 30 degrees a resistance that increased in a straight line to 10. Steeper slopes were assigned a resistance of 10, with slope calculated as the average within each 1-km cell. Alternative vegetation weights were compared because vegetation cover can influence cheetah habitat use (Verschueren et al. 2024), and different weighting schemes allowed a test of how strongly the modeled resistance surfaces depended on the vegetation weight (Zeller et al. 2012). The weights, slope thresholds, and fence multiplier were set by the author based on the cited literature, and were not fitted to cheetah movement data.

Three human-pressure weightings (the main weighting, equal, and infrastructure-emphasis), three vegetation weights (10, 20, and 40 percent), and five fence treatments (no fences, documented fences, near-barrier, Kruger documented, and Kruger conservative) were tested. Each alternative changed one setting from the main surface, giving a main surface plus eight alternatives. The first axis changed the balance among

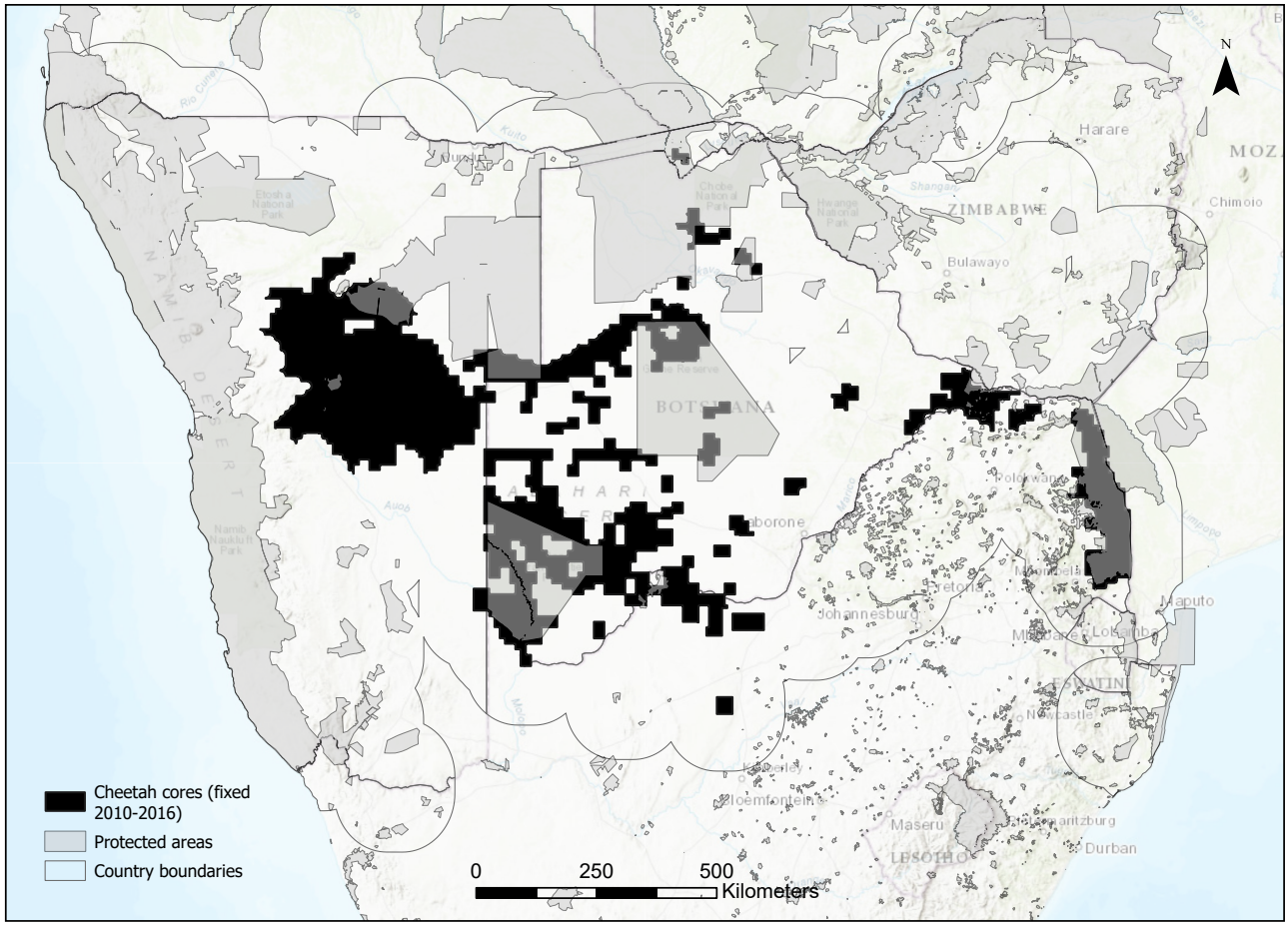


Fig. 1 Study extent, fixed historical range cores, and core groups used to define the modeled connectivity network. Core groups are an analysis construct and do not imply separate populations.

Table 1 Input layers, their data sources, how each was treated through time, and whether it entered the resistance surface.

Layer	Source	Treatment	In resistance	Role
Built-up fraction	Global Human Settlement Layer (GHSL), R2023A (Pesaresi 2023)	Dynamic	Yes	Human pressure
Nighttime radiance	Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime lights, VNL V2 (Elvidge et al. 2021); gas flares masked	Dynamic	Yes	Human pressure
Road density	Global Roads Inventory Project, version 4 (GRIP4) (Meijer et al. 2018)	Static	Yes	Human pressure
Livestock density	Gridded Livestock of the World, version 4 (GLW4), 2020 (Food and Agriculture Organization of the United Nations 2024; Gilbert et al. 2018); cattle, goats, and sheep combined	Static	Yes	Human pressure
Tree, non-tree, and bare cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B (DiMiceli et al. 2022)	Dynamic	Yes	Vegetation
Slope	Shuttle Radar Topography Mission (SRTM) (Farr et al. 2007)	Static	Yes	Terrain
Veterinary fences	World Wide Fund for Nature (WWF) Kavango–Zambezi Transfrontier Conservation Area (KAZA) fence map (World Wide Fund for Nature 2026); Kruger National Park fence references	Static; varied between scenarios	Yes, as a multiplier	Raises resistance where present; crossable, not a full barrier
Land cover	MODIS land cover, MCD12Q1 (Friedl and Sulla-Menashe 2022)	Context only	No	Masking and description; never differenced between years
Hydrography	HydroSHEDS rivers and lakes (Lehner and Grill 2013; Lehner et al. 2008; Messager et al. 2016)	Context only	No	Description and checks; water not treated as a barrier
Protected areas	World Database on Protected Areas (WDPA), August 2026 WDPA release (UNEP-WCMC and IUCN 2026)	Overlay	No	Reporting protected-area coverage only

Note: Dynamic layers change between the 2012, 2016, 2020, and 2024 snapshots; static layers are the same in every snapshot. Built-up area used the 2010, 2015, 2020, and 2025 products, nighttime lights the 2013, 2016, 2020, and 2024 products, and MODIS layers the snapshot years themselves. VNL = VIIRS nighttime lights; V2 = product version 2; MOD44B = MODIS vegetation continuous fields; MCD12Q1 = MODIS land cover type; HydroSHEDS = hydrological data and maps based on Shuttle Elevation Derivatives at multiple Scales. Component weights are given in the text.

the anthropogenic pressures, comparing the main weighting against an equal weighting and an infrastructure-emphasis weighting. The second changed how much the surface leaned on vegetation relative to anthropogenic pressure and terrain, at 10, 20, and 40 percent. The vegetation-balanced surface combined with the documented fence treatment carries the primary results, and the remaining combinations are reported as sensitivity rather than as alternative answers. This surface was chosen as the main one because it sits at the middle of the tested vegetation range and uses only fence locations that were documented. The predeclared protocol had treated all scenarios equally; a main surface was designated afterwards so that results could be reported once, with the alternatives as sensitivity. A link is only described as stable if it survives these alternatives (Section 2.9).

2.5 Connectivity modelling

For each core, a least-cost analysis in ArcGIS calculated the accumulated cost of moving through every reachable cell. The lowest accumulated cost was then used to draw paths from the starting core to each connected destination, for the 45 selected links (Section 2.6). Reusing the accumulated-cost calculation for several destinations produced the same set of pairwise paths with fewer repeat calculations. Circuitscape 5.17.1 in pairwise mode (Anantharaman et al. 2020) was used to model how current spread between each pair of cores. The Circuitscape analysis included all 253 core pairs in each of the four years, for a total of 1,012 pairwise comparisons.

2.6 Network structure

The 23 core areas were represented as fixed nodes across all four temporal snapshots (2012, 2016, 2020, and 2024). Each core was linked to its three nearest neighbors by Euclidean distance, and a minimum spanning tree, which is the smallest set of links needed to keep the core network connected, was added to ensure all cores were connected. This yielded 45 straight-line links, which defined the core pairs that received least-cost paths (Section 2.5).

Edge betweenness counts how many shortest paths between cores pass through a link. A high edge betweenness means that the link is important for holding the larger network together. The value was calculated on the 45-link network and divided by the 253 possible core pairs. To test how much the network design affected the rankings, the number of nearest neighbors was also varied from two to five.

2.7 Temporal comparison

The same core pairs, model assumptions, and 1-km grid were compared in 2012, 2016, 2020, and 2024. This kept the comparison consistent, so that the modeled differences came only from the input layers that changed through time. Absolute change was defined as the later value minus the earlier value. Percent change was defined as that difference divided by the earlier value, multiplied by 100. A negative change therefore means a connection became easier to cross in the model.

2.8 Protected-area coverage

The August 2026 WDPA release (UNEP-WCMC and IUCN 2026) was used to determine protected-area coverage, keeping terrestrial and coastal protected-area polygons, removing proposed areas (areas that are not officially protected), points, and marine-only areas, and merging overlaps. This left 2,301 records in the study extent. The protected-area data were the most recent available and were not reconstructed at each historical date. Cells inside the fixed cores were excluded, because the model forces current into and out of the cores, which would make protection look higher without describing the land between them. Cells with zero current were also excluded, because they could never enter a high-current group, so protected-area coverage was calculated only for areas that could carry modeled current between the cores. High-current cells were defined as those above the 80th, 90th, 95th, and 99th percentiles of modeled current, and their protected share was compared with that of all usable outside-core cells. No significance test is reported, because the result depended on the assumed size of the spatial blocks.

2.9 Stability and uncertainty

A link was treated as stable only if it passed three checks, and as uncertain if it failed one or more of them. First, under both alternative human-pressure weightings (equal and infrastructure-emphasis), at least 80 percent of the route had to stay within 5 km of the main route. Second, under the near-barrier fence treatment, the link's modeled cost could not rise by more than 20 percent, and at least 80 percent of its route had to stay within 5 km of the main route. Third, the direction of change from 2012 to 2024 had to stay the same under all three vegetation weightings. Separately, route location was checked from year to year by measuring how much of each least-cost path stayed within 1 km of the same path in the other snapshots.

2.10 Independent plausibility assessment

All inputs to the connectivity model are static. The cores are fixed from historical records, the landscape is represented by four snapshots, and the modeled paths are estimates rather than observed routes. To assess whether this static representation bears any relation to how cheetahs actually move, the model was compared with independent GPS tracking data from free-ranging cheetahs. This comparison is a plausibility check rather than a validation, because no resistance value was fitted to these data. The tracking data consist of 3,165 locations from nine free-ranging cheetahs (seven males and two females) collared near Thabazimbi, Limpopo Province, South Africa, between September 2003 and November 2008 (Marnewick et al. 2017a; Marnewick et al. 2017b). The tracked area lies inside the study extent but outside every core, so these animals were moving through the landscape between cores, which is the part of the map the model makes claims about. The data are also independent of the cores, as none of these collar locations appear in the range archive from which the cores were derived (Weise et al. 2017a).

Seventeen records without a complete date were removed, and the remaining locations were thinned to the earliest fix per

animal per calendar day, leaving 1,006 locations. Locations from the same animal on consecutive days were then paired into day-to-day moves. Moves shorter than 1 km were excluded because both ends fall within the same 1 km cell, which left 549 moves with a median distance of approximately 4 km. Each move was compared with 20 alternative moves the same animal could have made instead, following the logic of step-selection analysis, a method that compares observed movements with the alternatives available from the same starting point (Fortin et al. 2005; Thurfjell et al. 2014). Each alternative began at the same location as the real move, covered a distance drawn from that animal's own movements, and headed in a random direction. Because every alternative began where the real move began, the comparison reflects the animal's choice of destination rather than the general character of the area it lived in. Modeled resistance was read from the 2012 surfaces, the snapshot closest to the tracking period, at the end point of each move and averaged along the straight line between its start and end. For each move, the share of alternatives that reached higher resistance than the real move was recorded. If modeled resistance had no bearing on movement, this share would be approximately 0.50. Shares were averaged within each animal so that each of the nine animals counted once, and a 95% interval was derived by resampling the animals 10,000 times.

A broader comparison was also made between the 1,006 daily locations and ten random points per location drawn within the convex hull of all locations, repeated with the hull widened by 20 km. To see which inputs carried any result, the step comparison was repeated for each input layer on its own and for the main surface with each layer left out, and it was repeated on all nine resistance surfaces. The number of mapped roads crossed by each real move was also compared with its alternatives. These checks were added after the planned comparison was run, and none of them was used to change any weight or the choice of main surface. A weaker, non-independent comparison used the occurrence records of the range archive itself, thinned to one record per 10-km cell for 2010–2012 and for 2013–2016.

2.11 Monitoring prioritisation

For each of the 45 links, six measurements that could be used to prioritize monitoring were compiled and kept separate rather than combined into a single score, because no evidence-based way to weigh them against each other was available. The first was the edge betweenness of the link, which measures how important it is to the overall connectivity of the network. The second was the change in modeled resistance between 2012 and 2024, which indicates how much the landscape changed in terms of movement difficulty. The third, fourth, and fifth were the three stability checks (Section 2.9), which show how robust each link is to changes in model assumptions. The sixth was the straight-line length of the link, since longer links must cross more of the landscape. The protocol had also planned a monitoring index for every 10-km cell; that index was not built, so the monitoring-priority hypothesis could not be tested.

2.12 Software and reproducibility

ArcGIS Pro 3.7.2 (build 1901) and ArcGIS Python 3.13.13 were used for most of the spatial processing, Julia 1.12.7 and Circuitscape.jl 5.17.1 for the connectivity modeling, and Google Earth Engine to supply and align the satellite data. Most ArcGIS processing used Python scripts rather than ModelBuilder. An AI assistant (Claude, Anthropic) was used to help write and debug the analysis scripts and to draft the text, which the author reviewed, revised, and verified against the source outputs. The author takes full responsibility for the content.

3 Results

3.1 Connectivity change, 2012–2024

The modeled current concentration outside the cores was similar in 2012 and 2024, with a few small areas of change. Modeled current in 2024 is mapped in Fig. 2, and its change since 2012 in Fig. 3. The median change in modeled cost between cores was −1.224 percent, with 92 of all 253 core pairs becoming harder to move between and 161 becoming easier. The largest increase was +13.1 percent, for the link between cores 33 and 38. The overlap of the highest-current areas between successive snapshots, and between 2012 and 2024, was 0.91 to 0.93, where 1 would mean perfect overlap.

Of the 45 selected links, 18 became harder and 27 became easier. The overall pattern of change was not meaningfully related to baseline betweenness ($\rho \approx 0.01$), so the temporal-change hypothesis was mixed rather than supported. The 18 links that became harder did not have higher betweenness than the 27 that became easier (Mann–Whitney one-sided $p = 0.36$), and among the 18 harder links, the bigger losses did not fall on the more important links ($\rho = -0.22$, not significant). However, the losses were fairly concentrated, with the five hardest-hit of the 18 links carrying 46 percent of the total increase.

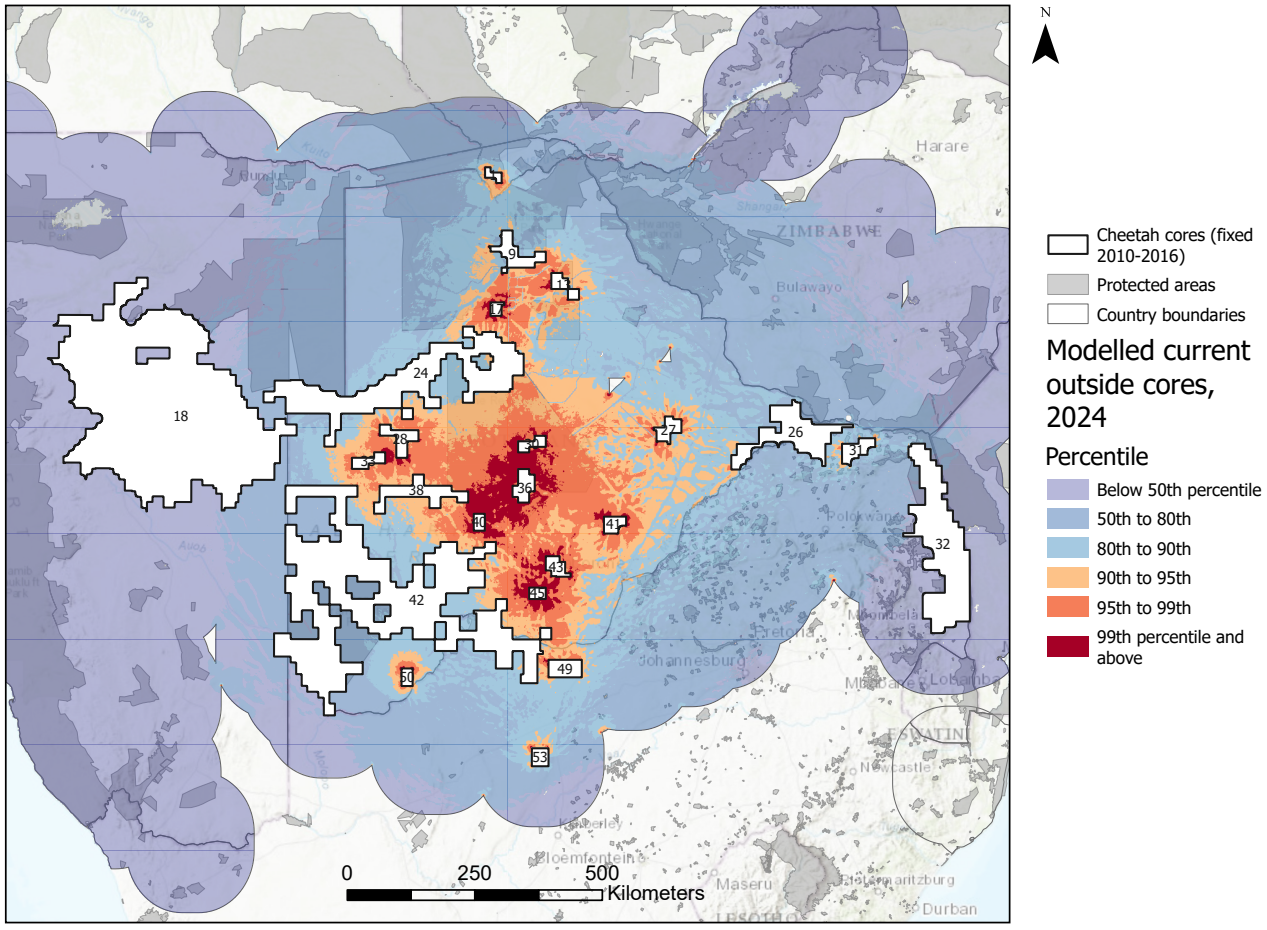


Fig. 2 Modeled current concentration for 2024, with the fixed core network shown for orientation.

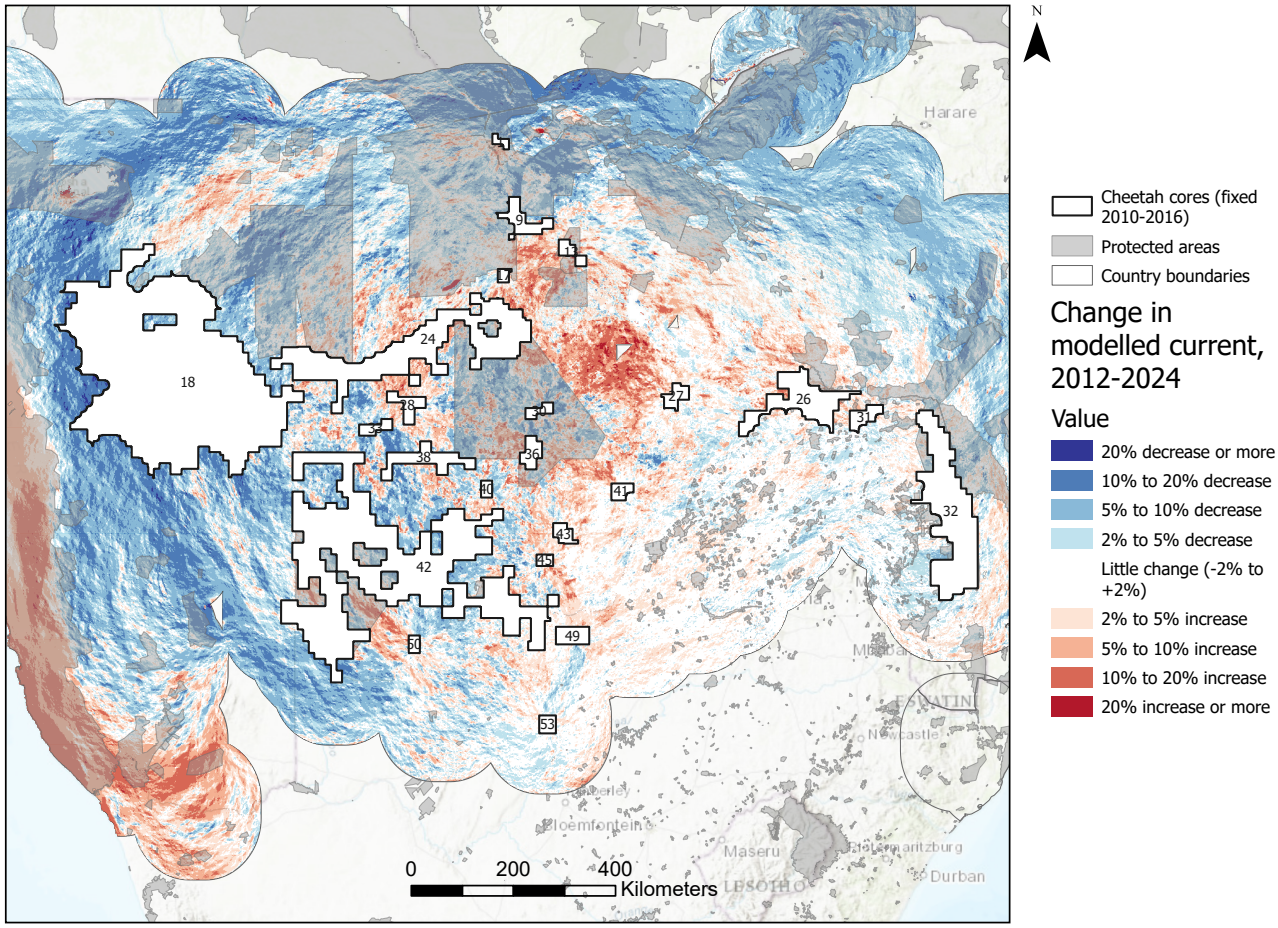


Fig. 3 Modeled current change between 2012 and 2024.

3.2 Priority links and sensitivity checks

Thirty-two of the 45 selected links passed the sensitivity tests and remained in the main priority group. The other 13 links were kept as uncertain links. The weighting test kept 34 links, and the fence test removed two more. The vegetation test did not remove any additional links from the remaining group. The main group is therefore not a small handful of links. It contains most of the network, which only partly matches the original expectation that the priorities would be limited to a small subset, so the link-stability hypothesis was only partly supported. The modeled links are shown with transportation and fence context in Fig. 4, and the 10 highest-ranked links are listed in Table 2.

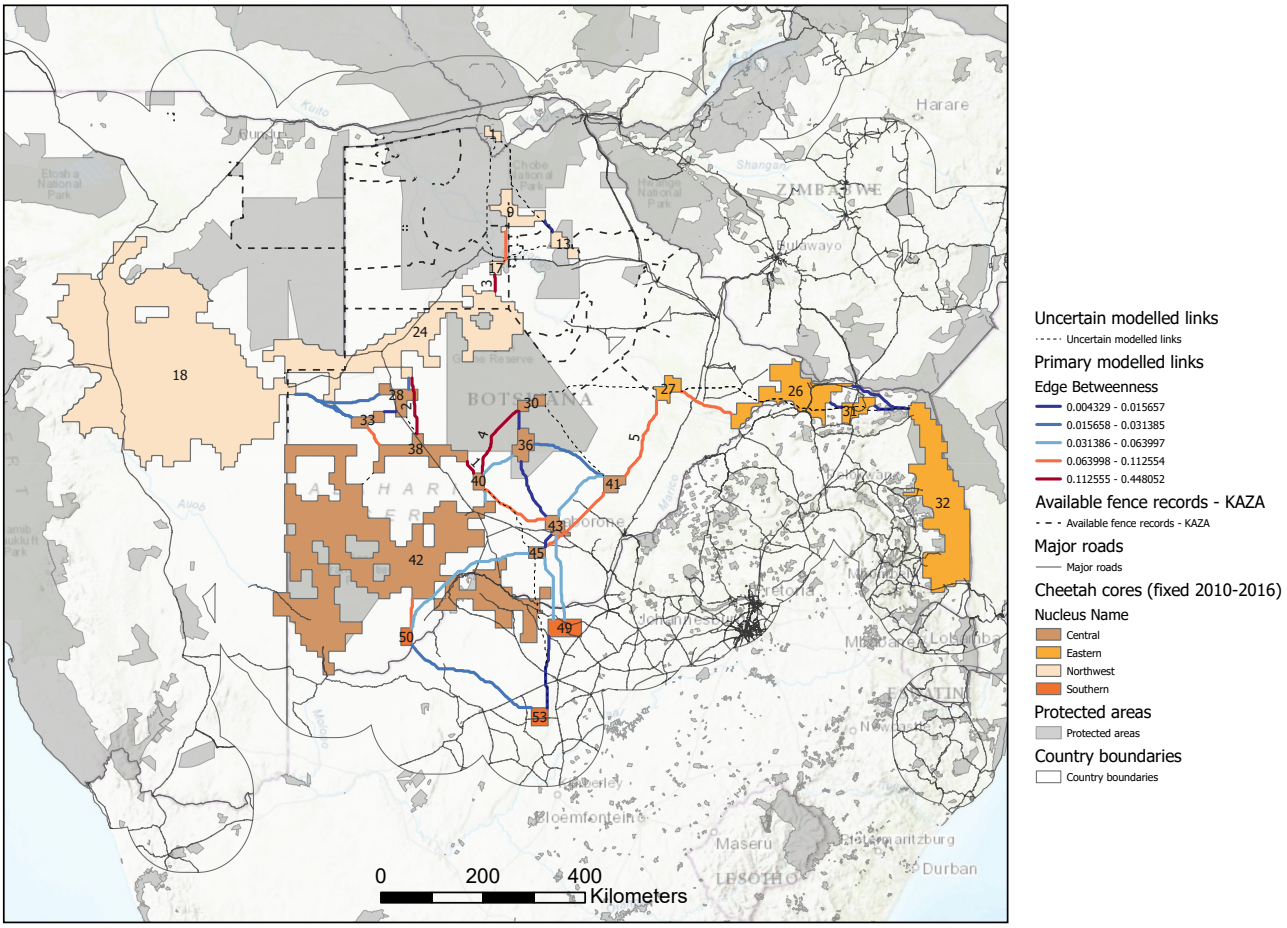


Fig. 4 Modeled core-to-core links with transportation and available fence context. Lines represent network links used to define least-cost analyses; they are not observed animal paths.

Table 2 Ten highest-ranked modeled priority links shown from the 32-link primary-priority set.

From	To	Rank	Betweenness	Type	Distance (km)	Change (%)
38	40	1	0.409	KNN+MST	120.6	+4.78
24	38	2	0.316	KNN	202.3	+7.44
17	24	3	0.300	KNN+MST	208.0	-5.78
30	40	4	0.238	KNN	169.2	+3.55
27	41	5	0.103	KNN+MST	196.8	-6.45
41	45	6	0.090	KNN	183.6	-5.50
33	38	7	0.079	KNN	108.1	+13.12
9	17	8	0.079	KNN+MST	109.0	-5.26
26	27	9	0.079	KNN+MST	221.8	-5.18
40	43	10	0.077	KNN	159.6	-1.56

Notes: The table shows the ten highest-ranked members of the 32-link primary-priority set; links are modeled structural network relationships, not observed animal routes. KNN = *k*-nearest-neighbor link; MST = minimum spanning tree link. "KNN+MST" identifies links retained by both network-construction rules.

The modeled cost of the 32 priority links was fairly stable, but the exact line taken by the least-cost path was less stable, meaning that the modeled route could shift by more than 1 km in some years. Of 180 route-and-year comparisons, 49 had less than 80 percent overlap within 1 km. This affected 21 of the 32 priority links. The route between cores 17 and 24 had no overlap in 2020. The vegetation assumptions changed the size of the modeled cost change, but not its direction for 43 of 45 links; the median absolute change grew from about 3.6 to 9.3 percent as vegetation was given more weight. Allowing two, three, four, or five nearest neighbors created 31, 45, 61, or 77 links, and the highest-ranked links changed. Link 17–24 was the only bridge under the three-neighbor choice, which means that removing it would disconnect the network.

3.3 Independent tracking check

Across 549 day-to-day moves, the location a cheetah actually reached had lower modeled resistance than most of the alternative moves it could have made (Table 3, panel A). On average, the real move had lower resistance than 56 percent of its alternatives (95 percent interval 51–61 percent), where 50 percent would indicate no difference, and seven of the nine animals showed this pattern. Averaged along the straight line between locations, the result was almost the same (55 percent). Modeled current showed no clear pattern: the real move had higher current than 53 percent of its alternatives, but the interval (47–59 percent) included 50, and only five of the nine animals leaned that way.

All nine resistance surfaces gave 55 to 56 percent (panel B). The five fence treatments gave identical values, because no mapped fence lies within 20 km of any track. Road density carried most of the result (panel C): on its own it reached 55 percent, and leaving it out reduced the main surface to 54 percent, with an interval that reached 50. Livestock, slope, and nighttime lights each showed a small lean of 51 to 52 percent. Built-up area barely varied in this district and could not separate real from alternative moves, and vegetation on its own leaned slightly in the opposite direction (48 percent, interval 43–52). Real moves crossed mapped roads less often than their alternatives (26 against 35 percent), and eight of the nine animals crossed fewer roads than expected (panel D).

In the broader comparison against random points across the tracked area (Table 4), cheetah locations had lower modeled resistance 60 percent of the time (seven of nine animals) and higher modeled current 65 percent of the time (six of nine animals). Widening the available area by 20 km gave the same picture.

In the weaker, non-independent comparison using the range archive itself, occupied 10-km cells numbered 413 in 2010–2012 and 450 in 2013–2016, with an overlap score of 0.239. Later records fell in slightly higher modeled resistance, and a larger share fell inside the fixed cores. Almost no records were close to the priority paths (zero early cells and three later cells). No occurrence records were available after 2016, so this comparison cannot speak to the 2020 or 2024 results.

The monitoring-priority hypothesis was not tested, because the cell-level monitoring index it depended on was not built (Section 2.11).

Table 3 Day-to-day moves of nine free-ranging cheetahs (Limpopo, 2003–2008) compared with moves of the same length the animals could have made instead, on the 2012 surfaces. Panels show the main surface, all nine surfaces, single input layers, and road crossings.

Comparison	Share of alternatives the real move beat	95% interval (resampling animals)	Animals above 0.50
A. Main surface, 549 day-to-day moves, 20 alternatives each			
Resistance at the end point (lower expected)	0.56	0.51–0.61	7 of 9
Resistance along the straight-line path	0.55	0.51–0.59	7 of 9
Current at the end point (higher expected)	0.53	0.47–0.59	5 of 9
Current along the straight-line path	0.52	0.47–0.58	5 of 9
B. All nine surfaces, resistance at the end point			
Main surface	0.56	0.51–0.61	7 of 9
Equal human-pressure weights	0.56	0.51–0.61	7 of 9
Infrastructure emphasis	0.56	0.50–0.61	6 of 9
Vegetation weight 10%	0.56	0.50–0.61	7 of 9
Vegetation weight 40%	0.55	0.51–0.60	6 of 9
Four other fence treatments[a]	0.56	0.51–0.61	7 of 9
C. Single input layers, resistance at the end point			
Road density	0.55	0.50–0.62	6 of 9
Livestock	0.52	0.49–0.56	5 of 9
Terrain (slope)	0.52	0.50–0.54	4 of 9
Night lights	0.51	0.51–0.52	8 of 9
Built-up area[b]	0.50	0.50–0.50	3 of 9
Vegetation	0.48	0.43–0.52	3 of 9
Main surface without road density	0.54	0.50–0.57	7 of 9
D. Road crossings along the straight-line path (fewer expected)			
Any mapped road (26% of real moves crossed, 35% of alternatives)	0.54	0.52–0.57	8 of 9
Major roads only (1% against 2%)	0.51	0.50–0.51	5 of 9

Note: Each real day-to-day move of at least 1 km is compared with 20 moves the same cheetah could have made instead, with the same start, a step length drawn from that animal's own moves, and a random direction. The score is the share of those alternatives the real move beat; 0.50 means no difference. Each animal counts once, and the 95% interval comes from resampling the nine animals. [a] No mapped fence lies within 20 km of any track, so all five fence treatments give identical values; these data cannot test the fence component of the model. [b] Built-up area barely varies in this district, so it cannot separate real and alternative moves.

Table 4 Modeled resistance and current at daily locations of nine free-ranging cheetahs (Limpopo, 2003–2008) compared with random available points from the same landscape, on the 2012 surfaces. This is the broader, less strict comparison.

Surface	Measure	Cheetah fixes (median)	Available (median)	Expected direction (share)	20-km buffer (share)	Animals (of 9)
Final model (vegetation-balanced, documented fences)	Resistance (lower expected)	2.23	2.45	0.60	0.60	7
	Current (higher expected)	0.156	0.145	0.65	0.62	6
Earlier model (pressure and terrain only, no fences)	Resistance (lower expected)	2.03	2.34	0.61	0.61	–
	Current (higher expected)	0.143	0.128	0.65	0.63	–

Note: The comparison uses 1,006 daily locations from nine free-ranging cheetahs (2003–2008) against 10,060 random points from the same landscape, on the 2012 surfaces. "Expected direction (share)" is the probability that a random cheetah location had lower resistance (or higher current) than a random available point; 0.50 means no difference. "20-km buffer" repeats this with a wider available area. "Animals" counts how many of the nine had a median in the expected direction. No location fell inside a core. In the stricter step comparison (Table 3), the resistance result held but the current result did not.

3.4 Protected-area coverage of priority connectivity

Of the usable outside-core area, 31.26 percent was protected. In 2024, only 23.27 percent of the area above the 80th percentile of modeled current was protected, rising to 25.54 percent above the 90th percentile, 33.45 percent above the 95th, and 44.63 percent above the 99th (Table 5). The same general pattern appeared in all four years; in 2012 the corresponding values were 24.23, 26.31, 35.57, and 44.17 percent. High-current land outside the cores was therefore less protected than the comparison area up to the 90th percentile and more protected above the 95th. Because the direction changed across thresholds, the protected-area-coverage hypothesis was not supported under its full rule.

Table 5 Protected-area coverage of modeled high-current classes.

Year	Current percentile	Cells	Protected (%)	Background (%)	Difference (pp)
2012	80	418,295	24.23	31.26	-7.03
2012	90	209,148	26.31	31.26	-4.95
2012	95	104,574	35.57	31.26	+4.31
2012	99	20,915	44.17	31.26	+12.92
2024	80	418,292	23.27	31.26	-7.99
2024	90	209,146	25.54	31.26	-5.72
2024	95	104,573	33.45	31.26	+2.19
2024	99	20,915	44.63	31.26	+13.37

Note: Values use cells outside the fixed cores with positive modeled current. Background is the protected share of all such cells; differences are in percentage points. No permutation p-value is reported, because inference was sensitive to the assumed spatial block size.

4 Discussion

4.1 Connectivity change and comparison with previous work

The modeled ease of movement between cores changed only slightly from 2012 to 2024. Across all 253 core pairs, the median cost fell by about 1.2 percent, and individual connections changed in both directions rather than declining together. These results describe landscape structure and do not show where cheetahs actually moved. The changes that did occur were concentrated in a minority of links, with five links carrying almost half of the total increase in modeled cost, but they did not fall disproportionately on the links most important to the network. Because the cores were held fixed and most input layers were static, any modeled change came from the three time-varying layers: built-up area, nighttime lights, and vegetation cover. The overall picture is therefore one of mixed change rather than a broad decline.

The general approach used here resembles regional connectivity studies of elephants and of several species across the Kavango–Zambezi Transfrontier Conservation Area (Brennan et al. 2020; Naidoo et al. 2024), and work showing how veterinary fences alter large-mammal movement (Osipova et al. 2018). Similar mapping tools can be applied to different species, but the resistance values themselves cannot simply be transferred, because each species responds differently to roads, vegetation, fences, and people. Veterinary fences are a clear example. In northern Botswana many fence sections are already in disrepair, and a recent disease-risk assessment found that removing several sections would not raise the risk of the main transboundary livestock diseases, prompting consultations on opening low-risk sections (Rosen et al. 2026). This supports treating fences as raising resistance rather than as complete barriers, as in the present model, and it makes the fence scenarios relevant to a current policy question. The model does not, however, show what removing a particular fence would mean for cheetahs, because no fence-removal scenario was run.

4.2 Protected areas and conservation context

The strongest concentrations of modeled current often reached protected areas, but the broader land connecting them was less protected than the study area as a whole. This suggests that maintaining modeled connectivity between cheetah range cores depends on farms, communal land, and other land outside formal protected areas, not only on parks. The pattern does not mean that protected areas have failed. Parks are often located on land of lower agricultural value, which may partly shape where both protection and modeled current occur. The result is also descriptive: the significance of the difference depended on the assumed size of the spatial blocks, so it should be read as a pattern rather than a tested effect.

4.3 Priority links and field-survey needs

Links that stayed stable under several assumptions are the strongest candidates for conservation attention. Thirty-two of the 45 links met that standard, which is most of the network rather than the small subset originally expected, so the stability screen narrows priorities less than intended. Agreement between scenarios is also not complete proof, because the scenarios share many of the same input layers. The 13 uncertain links are better treated as places where field information could change the result. No combined score ranks individual survey cells, so an organization could use these links to choose general survey areas and then adjust the order by access, cost, safety, and local knowledge. Link 17–24 is a clear first candidate: it was the only link holding two parts of the three-neighbor network together, yet its mapped route was highly unstable and had no overlap in 2020.

4.4 How far the tracking check supports the model

In one district, free-ranging cheetahs moving from one day to the next tended to reach ground that the model rates as easier than the ground they could have reached instead. This held for seven of the nine animals and on all nine resistance surfaces. The check could have failed: had the animals chosen higher-resistance ground, the result would have pointed the other way. The effect was modest, however, with real moves beating about

56 percent of their alternatives against 50 percent by chance, so the result is consistent with the model rather than a close match.

Most of the signal came from roads. Road density alone performed about as well as the full surface, and the animals crossed mapped roads less often than their alternative moves did, in line with the road avoidance reported for GPS-collared cheetahs in the Maasai Mara (Klaassen and Broekhuis 2018). Vegetation on its own leaned slightly the wrong way, and modeled current showed no clear pattern. Cheetahs in this ranchland may use denser bush for cover and hunting, and the tracks predate the 2012 vegetation layer; either could explain the vegetation result, but neither was tested. No weight was changed in response to this check, which keeps it independent of the model. The check does not show that the model holds elsewhere, because it covers nine animals in one part of South Africa, and it says nothing about change over time, because the tracks end in 2008. It also could not test fences, because none are mapped near the tracks and the private game fences in the area are not in the model. The planned comparison with camera-trap data from central-eastern Namibia (Verschueren et al. 2024) was not carried out, and independent movement data from fenced and more developed areas would be the most useful next test.

4.5 Limitations

The model shows where movement may be easier or harder, but it does not show where cheetahs actually traveled. The cores and several landscape layers were held constant through time, which allowed a fair comparison between years but limited what could change. The resistance values were set by the author, guided by published studies, and were not fitted to cheetah movement data; rainfall was not checked, so some vegetation change may reflect wet or dry years rather than land use, and source years did not always match the snapshot years. The mapped paths are broad planning areas rather than exact travel routes, and the only independent check came from nine animals in one district before 2012, which could not test the fence component and found a lean carried mostly by roads. Longer-running tracking datasets exist for other parts of the network, such as the Okavango Delta and central Namibia, but they are held privately and were not requested within this project, so the results should be tested further before use at another scale or in another region.

5 Conclusion

Modeled connectivity between fixed southern African cheetah range cores changed only slightly between 2012 and 2024, with individual links changing in both directions rather than declining together. The strongest concentrations of modeled current often reached protected areas, but the wider connecting landscape was less protected than the study area as a whole. Thirty-two links were retained as main priorities, and 13 were identified as places where more field information is needed. These are modeled landscape connections, not observed cheetah routes, and they are best used to decide where field monitoring should begin.

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